Chef Attribution from Recipe Text with DistilBERT

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1 Introduction

We fine-tune `distilbert-base-uncased` [Sanh et al., 2019] with the Hugging Face Transformers toolkit [Wolf et al., 2020] to predict the originating chef of a recipe. The dataset holds 2,999 labelled recipes; 14 entries are perfect duplicates, so we work with 2,985 unique samples spanning six chefs (365–801 recipes each). The provided baselines TF-IDF on the description alone and on all fields score 30.0% and 43.0% accuracy. Our final model reaches 92.13% validation accuracy and macro-F1 0.916 on an 80/20 stratified split, while keeping inference feasible on standard laptops.

2 Models

We turn every recipe into a single string in the order *name* $\rightarrow$ *ingredients* $\rightarrow$ *tags* $\rightarrow$ *description* $\rightarrow$ *steps*. This layout keeps the short, informative parts at the front so that any truncation affects the long step lists first. The same concatenation is used for training, validation, and test prediction, and the overall flow is shown in Figure 1.

The classifier is the standard DistilBERT encoder with a 768$\rightarrow$768 pre-classifier, a ReLU activation, a single dropout layer with $p = 0.2$, and a final linear layer that produces six logits. We rely on the Hugging Face implementation for both the model and the fast uncased tokenizer. Dynamic padding keeps each batch as short as possible.

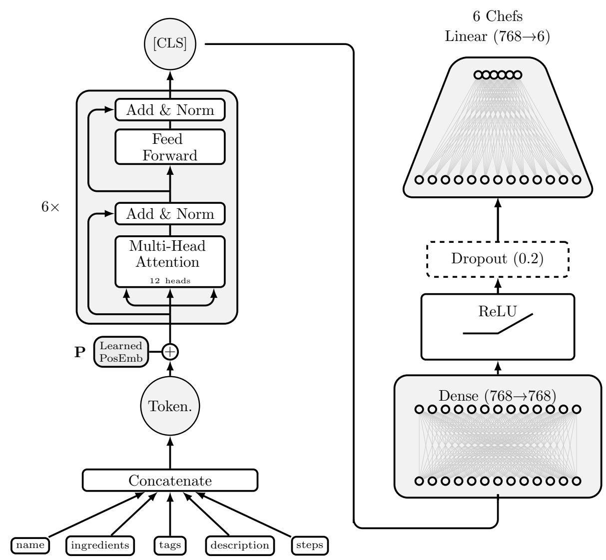

Figure 1: Data flow from raw fields to chef prediction.

3 Experimental Setup

3.1 Datasets

After removing 14 duplicate texts we work with 2,985 unique recipes. Stratifying an 80/20 split with seed 42 yields 2,388 training and 597 validation samples, and the blind test file contains 823 items. When we tokenize the deduplicated corpus before any truncation, the median sequence length is 234 WordPiece tokens, the 95th percentile is 418, the maximum is 1,065, and 98.7%

of examples remain within 512 tokens, so we keep DistilBERT's default maximum length.

3.2 Metrics

Accuracy is the primary metric to match the grading rubric. We also track macro-F1, per-class precision, and per-class recall to ensure the minority chefs are not ignored.

3.3 Parameters/Hyperparameters

Training uses AdamW with learning rate 2×10^{-5}, weight decay 0.01, gradient clipping at 1.0, and a linear warm-up over 6% of the steps. Batch sizes are 8 for training and 16 for evaluation with dynamic padding. We evaluate every 100 optimisation steps and stop after two non-improving macro-F1 scores, keeping the best checkpoint. All random seeds are set to 42.

All scripts, configuration files, and experiment artefacts are version-controlled in our Git repository.

4 Results

The best checkpoint (training step 1,300) reaches 92.13% validation accuracy, macro-F1 0.916, and training loss 0.45. Every class has precision above 0.85 and recall in $[0.862, 0.972]$. Appendix Figure 3 plots the training and validation curves, while the normalised confusion matrix in Appendix Figure 4 shows the minority chefs (1533 and 6357) are rarely mislabelled as the majority chef (4470). On the blind test set, predicted class proportions stay within ± 2.3 percentage points of the training priors, suggesting the model generalises rather than memorising class frequencies.

5 Discussion

Manual review of validation and test predictions highlights recurring cues: health-conscious wording (“diabetic”, “fat-free”) tags chef 5060, “once-a-month” freezer phrases map to chef 3288, and concise single-serving recipes point to chef 6357. Mistakes usually involve very short descriptions, cross-over dishes shared by multiple chefs, or the 1.8% of items truncated past 512 tokens. Deduplicating before splitting prevented label leakage that otherwise inflated accuracy by several points. Switching the classifier activation to GELU or widening the head did not improve macro-F1, so we kept the default architecture.

6 Future Work

Next steps include testing class-weighted loss or simple calibration to tighten recall for chefs 1533 and 3288, running cross-validation to estimate variance beyond a single stratified split, and applying attention or gradient-based attribution to confirm the model exploits stylistic cues rather than only explicit keywords.

Bibliography

References

[Sanh et al., 2019] Sanh, V., Debut, L., Chaumond, J., and Wolf, T. (2019). Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. In *NeurIPS 2019 Workshop on Energy Efficient Machine Learning and Cognitive Computing*.

[Wolf et al., 2020] Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., and Brew, J. (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of the*

2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pages 38–45. Association for Computational Linguistics.

Appendix: Supporting Figures

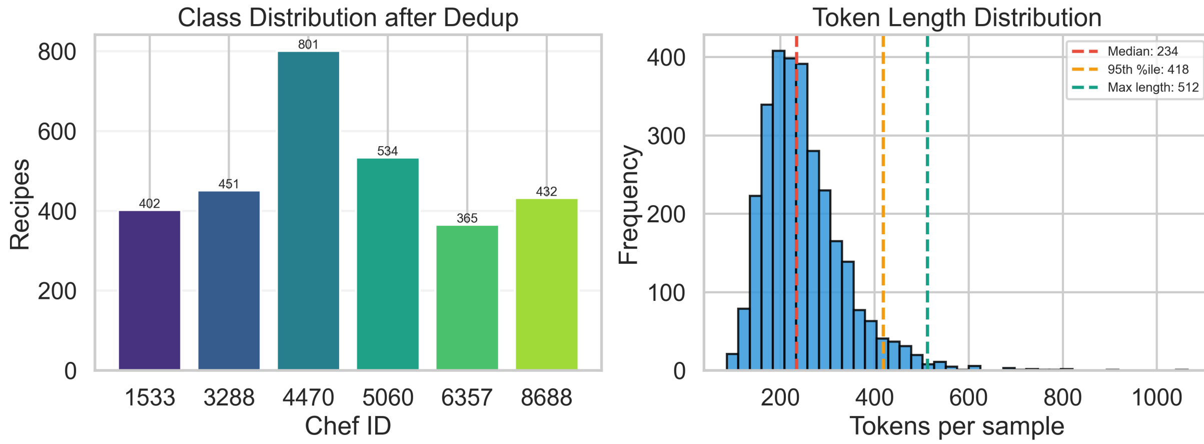

Figure 2: Class distribution and token lengths.

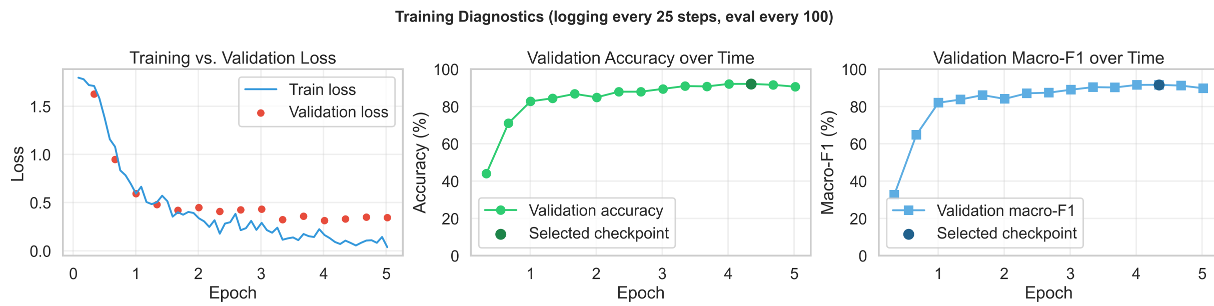

Figure 3: Training metrics over time.

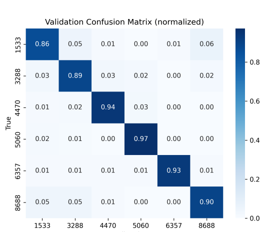

Figure 4: Validation confusion matrix (normalized).